Assignment No	4
Title	Stacks
Objective	Array Implementation of Stacks
Roll No	MCA2649

Source Code:

```
//Name:- Satish Rathod
//Roll No:- 2649
//Module 3:- Stacks
//Array Implementation of Stacks

import java.util.Scanner;

class Stack
{
   int n;
   int tos;
   int[] arr;

   public Stack(int size)
   {
      n = size;
      arr = new int[n];
      tos = -1;
   }
```

```
// Push
public void Push(int x)
{
   if(tos == n - 1)
   {
      System.out.println("Stack Overflow");
      return;
   }

   tos++;
   arr[tos] = x;
}

// Pop
public void Pop()
{
   if(tos == -1)
   {
      System.out.println("Stack Underflow!");
      return;
   }

   System.out.println("Element popped is: " + arr[tos]);
   tos--;
}

// Peek
public void Peek()
{
   if(tos == -1)
```

```
    {
      System.out.println("Stack Underflow");
      return;
    }

    System.out.println("Element at the TOS: " + arr[tos]);
  }

  // Display
  public void Display()
  {
    if(tos == -1)
    {
      System.out.println("Stack Underflow");
      return;
    }

    System.out.println("Stack Elements:");

    for(int i = tos; i >= 0; i--)
    {
      System.out.println(arr[i]);
    }
  }
}
//end of class Stack


class AStack
{
```

```
public static void main(String[] args)
{
   Scanner sc = new Scanner(System.in);

   Stack s = new Stack(5);

   int choice, d;

   do
   {
      System.out.println("\n*** Stack Array Implementation ***\n");

      System.out.println("1. Push an element on the stack");
      System.out.println("2. Pop an element from the stack");
      System.out.println("3. Peek at the stack");
      System.out.println("4. Display the stack");
      System.out.println("5. Exit\n");

      System.out.print("Enter your choice: ");
      choice = sc.nextInt();

      switch(choice)
      {
         case 1:
                  System.out.print("Enter element to push: ");
                  d = sc.nextInt();
                  s.Push(d);
                  break;

         case 2:
```

```
                s.Pop();
                break;

            case 3:
                s.Peek();
                break;

            case 4:
                s.Display();
                break;

            case 5:
                System.out.println("Exiting...");
                System.exit(1);

            default:
                System.out.println("Incorrect choice!");
                break;
        }//end of switch

    } while(choice != 5);  //end of do while


  }//end of main function
}
//end of class AStack
```

Output:

1.Push

```
D:\MCA2649\ADS2649>java AStack

*** Stack Array Implementation ***

1. Push an element on the stack
2. Pop an element from the stack
3. Peek at the stack
4. Display the stack
5. Exit

Enter your choice: 1
Enter element to push: 10
```

2.Pop

```
*** Stack Array Implementation ***

1. Push an element on the stack
2. Pop an element from the stack
3. Peek at the stack
4. Display the stack
5. Exit

Enter your choice: 2
Element popped is: 50
```

3.Peek

```
*** Stack Array Implementation ***

1. Push an element on the stack
2. Pop an element from the stack
3. Peek at the stack
4. Display the stack
5. Exit

Enter your choice: 3
Element at the TOS: 50
```

4.Display

```
*** Stack Array Implementation ***

1. Push an element on the stack
2. Pop an element from the stack
3. Peek at the stack
4. Display the stack
5. Exit

Enter your choice: 4
Stack Elements:
50
40
30
20
10
```

5.Exit

```
*** Stack Array Implementation ***

1. Push an element on the stack
2. Pop an element from the stack
3. Peek at the stack
4. Display the stack
5. Exit

Enter your choice: 5
Exiting...
```

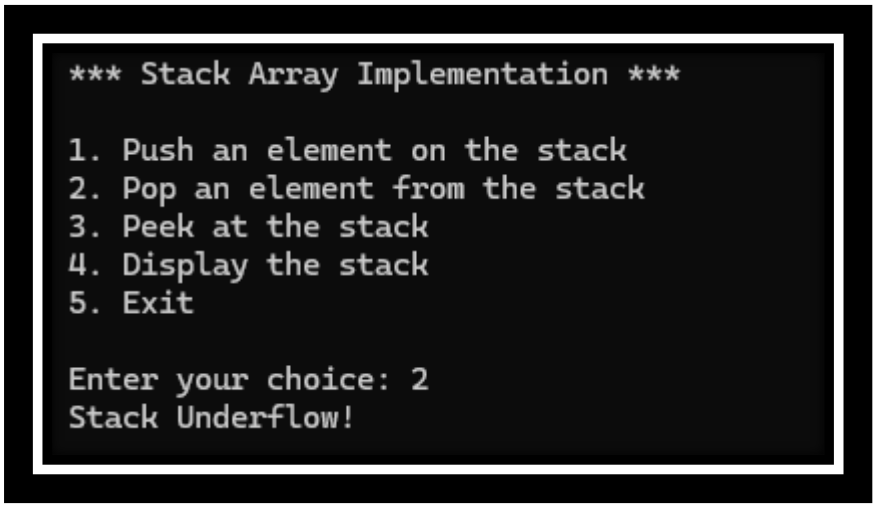
6.OverFlow

```
*** Stack Array Implementation ***

1. Push an element on the stack
2. Pop an element from the stack
3. Peek at the stack
4. Display the stack
5. Exit

Enter your choice: 1
Enter element to push: 60
Stack Overflow
```

7.UnderFlow


*** Stack Array Implementation ***
1. Push an element on the stack
2. Pop an element from the stack
3. Peek at the stack
4. Display the stack
5. Exit
Enter your choice: 2
Stack Underflow!